

Introduction to Machine Learning

Understanding Assessment

60 minutes | 50 points

Question 1: How learning happens

[10 points]

Describe the steps involved in training a supervised machine-learning model and explain the purpose of each step.

Hint: Imagine you are given a house-price dataset and asked to build a model that predicts the price of a new house. What would you do?

Question 2: Model values and settings

[8 points]

Consider the following model:

```
model = MLPClassifier(  
    hidden_layer_sizes=(32, 16),  
    learning_rate_init=0.001  
)  
model.fit(X_train, y_train)
```

Choose from: gradients, activations, hyperparameters, features, parameters.

After training, `coefs_` and `intercepts_` are _____. The layer sizes and learning rate are _____. Fill in the blanks and explain the role of one item from each group.

Question 3: Two prediction tasks

[8 points]

System A predicts a house's price. System B produces ten scores for a handwritten image and chooses the digit with the largest score.

Choose from: linear model, MLP, classification, neuron, regression.

System A is a _____ task. System B is a _____ task. Fill in the blanks and explain what the model output and prediction error mean in each case.

Question 4: Residuals and mean squared error

[10 points]

A linear model uses

$$\hat{y} = wx + b.$$

For two examples, suppose

$$x = [1, 3], \quad w = 2, \quad b = 1, \quad y = [2, 8].$$

- (i) Calculate the two predictions $\hat{y}$.
- (ii) Calculate the residual for each prediction.
- (iii) Calculate the mean squared error (MSE).
- (iv) Is the following statement true or false? Explain your answer.

“The average residual is zero, so the model made no mistakes.”

Question 5: Complete MLP forward pass**[14 points]**

A neural network has two inputs, two hidden layers, and two output scores. Treat all vectors as row vectors. Its input and parameters are

$$x = [1, 2], \quad W_1 = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}, \quad b_1 = [0, 1],$$

$$W_2 = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}, \quad b_2 = [0, 0],$$

$$W_3 = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}, \quad b_3 = [0, 0].$$

Perform the complete forward pass using ReLU after both hidden layers. Show every value before and after ReLU, give the final class scores and predicted class, and identify the learned parameters.